

A Mini-Project 1 Report on

EXPLAINABLE AI-BASED SCREENING OF RETINOBLASTOMA

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**Submitted to the Department of Computer Science & Engineering, GNITS in the
partial fulfillment of the academic requirement for the award of B.Tech (CSE)
under JNTUH**

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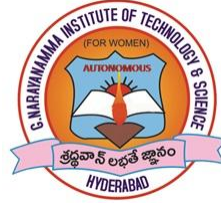


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Certificate

This is to certify that the Mini-Project 1 report on “**Explainable AI-Based Screening of Retinoblastoma**” is a bonafide work carried out by **A. Hasini (24251A05T1), D. Sri Nithya (24251A05T5), S. Sri Nithya (24251A05V5)** in the partial fulfillment for the award of B.Tech degree in Computer Science & Engineering, G. Narayanamma Institute of Technology & Science, Shaikpet, Hyderabad, affiliated to JNTUH, Hyderabad under our guidance and supervision.

The results embodied in the project work have not been submitted to any other University or Institute for the award of any degree or diploma.

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This is to certify that **A. Hasini (24251A05T1), D. Sri Nithya (24251A05T5), S. Sri Nithya (24251A05V5)** II B.Tech, has successfully completed project work in “**CoE for Deep Learning in Eye Disease Prognosis**”, CSE Department as part of Grant-in Aid under DST/DSIR in collaboration with Aravind Eye Hospital, Hyderabad.

The project titled “**Explainable AI-Based Screening of Retinoblastoma**” that is being submitted in partial fulfilment for the award of B. Tech, Computer Science and Engineering, G. Narayanamma Institute of Technology & Science affiliated to JNTUH, Hyderabad, is a record of bonafide work carried out by them in our guidance and supervision.

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Abstract

Retinoblastoma is a rare but life-threatening eye cancer that mainly affects children, and early diagnosis is crucial to prevent vision loss and improve survival rates. This project presents an Explainable Artificial Intelligence (XAI) based system for the early detection of retinoblastoma using retinal fundus images. A deep learning model is designed to analyze fundus images and classify them into normal and retinoblastoma-affected categories.

The system aims to provide accurate and reliable predictions to assist in early screening and diagnosis of pediatric ocular cancer. To improve transparency and trust in AI-assisted medical diagnosis, explainability techniques such as LIME and SHAP are applied to interpret the predictions generated by the deep learning model. These techniques produce visual explanation maps that highlight important retinal regions influencing the classification decision. Such interpretability helps in understanding the model's decision-making process and supports clinicians in validating AI results. The proposed system demonstrates effective performance along with meaningful visual explanations, making it suitable for clinical decision support and early retinoblastoma screening

Keywords: Retinoblastoma, Retinal Fundus Images, Convolutional Neural Network, Transfer Learning, MobileNetV2, LIME, SHAP, Deep Learning

Sustainable Development Goals (SDGs): (SDG 3): Good Health and Well-Being, (SDG 9): Industry, Innovation, and Infrastructure.

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1. INTRODUCTION

1.1 Background of the study

Retinoblastoma isn't a disease you hear about every day, but for small children, it's a really serious form of eye cancer. Most of the time, it hits kids who are under five years old. The biggest challenge here is that toddlers can't really explain if something is wrong with their vision. Parents might notice a weird white glow in their child's eye—kind of like a "cat's eye" in a photo or maybe a slight squint, but they often brush it off as nothing.

The situation is even tougher in rural areas or small towns. Regular eye check-ups just aren't common there, and most local hospitals don't have specialized eye doctors. This leads to a huge problem: the cancer gets caught way too late. Missing the early window, treating it becomes much harder, and the chance of saving the child's sight drops significantly.

This is where technology could actually step in and help. Artificial Intelligence (AI) doing amazing things in medicine lately, mostly because it can spot tiny patterns in medical images that even a trained human eye might miss. For the project, retinal fundus images are used - these are basically detailed photos of the back of the eye.

But there's a catch. If a computer simply says "Cancer" or "Normal" without explaining why, doctors cannot rely on the prediction alone. In medicine, trust is essential. So, the system was designed not only to make predictions but also to include Explainable AI (XAI). This allows the model to “show its work” by highlighting the specific parts of the eye image that influenced its decision. By making the AI explain its predictions, the system becomes more reliable and useful for helping doctors detect the disease at an early stage.

1.2 Problem Statement

Early detection of retinoblastoma is extremely important, but it is often challenging because children may not show the same symptoms in the initial stages, and periodic eye examinations are not routinely performed in many underserved or remote areas. Diagnosis usually depends on a comprehensive clinical examination and the experience of specialists, which can be time-consuming and resource-intensive, and early signs of the disease may be missed. The shortage of trained ophthalmic specialists in many regions further delays diagnosis and treatment. Although artificial intelligence models can assist in detection, many operate as “black boxes,” providing predictions without explaining the reasoning behind them. In medicine, this lack of transparency is a major concern because clinicians require understandable and trustworthy evidence before relying on automated systems. Therefore, an effective solution should not only provide accurate predictions but also offer clear and interpretable explanations for those predictions.

1.3 Existing Systems

At present, the only way to check for retinoblastoma is by the retinologist checking it with an instrument called the ophthalmoscope, but the retinologist ultimately depends upon the doctor's ability and observation skills to make a determination of the presence of retinoblastoma. In larger institutions, some very sophisticated imaging technologies are utilized, such as fundus photography, that will allow the retinologist to obtain detailed images of the retina that can be stored and easily retrieved later. The retinologist may recommend an MRI or CT scan if he/she would like to have a better understanding of whether retinoblastoma has spread (in cases of suspected retinoblastoma). MRI and CT scans are very effective for determining the degree of involvement of the retina; however, they can be quite costly and not offered at all hospitals. In addition to the use of manual systems, there have been several recently developed computer-based systems for assisting in diagnosing retinoblastoma. These systems utilize machine or deep learning techniques to analyze retinal images and then classify them as either normal or abnormal after evaluating many sample images. Utilizing this approach will not only reduce the workload of the retinologist but also help expedite the diagnosis of retinoblastoma. There are many systems utilizing Convolutional Neural Networks (CNN's), which have shown to be effective for image-related tasks.

Unfortunately, the majority of these systems provide no explanation of how they arrived at the result; they just provide the result itself, which creates a problem of trust. When a physician is providing medical care to their patient, they cannot rely on a system to only provide the answer without any rationale. Many times, the current systems are trained using very small or specific datasets. This creates a problem when new or different images are presented to them, even if they perform well for certain images. In reality, data is considerably less organized than the data on which many current solutions are based. Many of the current systems are also not user-friendly and require a significant degree of technical knowledge or set-up, which is often not found in all hospitals, particularly in rural hospitals. Overall, while current systems are useful, there are still significant gaps. They are either over-reliant on human intelligence or, in the case of A.I., lack sufficient explanation or flexibility.

1.4 Drawbacks of Existing System

Manual diagnosis depends on the doctor's knowledge and expertise. Therefore, if the doctor has seen numerous cases like this, they can identify the disease sooner rather than later. However, in some instances (especially early stage), the signs may be so subtle that they can be easily missed. Thus, patients who have delayed diagnoses must also endure much more difficult treatment options.

Advanced diagnostic procedures such as scanning/image technology are often prohibitively expensive. Many hospitals (primarily in rural or small areas) do not possess the necessary resources to provide these types of diagnostic facilities for their patients. As a result, patients often have not had an accurate diagnosis at the appropriate time. Often, this process can be time intensive. Patients may have to travel to many different facilities or to see multiple specialists before being able to receive an accurate diagnosis. As a result, the timing of the diagnosis can also significantly impact the prognosis for the patient.

With regards to AI-based systems, the vast majority of these systems function as "black boxes." That is, they return an output value; however, they do not disclose how they derived that output. This is especially problematic within the medical field because physicians need to have high confidence levels before making decisions about their patients. If an AI-based system does not provide next-level transparency, physicians will often not trust using that system as their sole means for making a decision. Also, some systems are trained on fewer data points, which can affect their performance as a result. If an image input looks just a little different than the images used to train the model, it may predict incorrectly. Finally, the usability of some systems can also be a problem; often, they require technical skills to operate, making them unsuitable for many different types of personnel in the healthcare world. So, although these systems can help with early detection and making decisions, they cannot yet be considered completely trustworthy.

1.5 Proposed System

A system is proposed that combines deep learning and Explainable Artificial Intelligence (XAI) techniques for the early detection of retinoblastoma using retinal fundus images. The images undergo preprocessing steps such as enhancement, normalization, and noise reduction to improve quality and ensure accurate analysis. A deep learning model is trained on labeled retinal images to identify patterns associated with retinoblastoma and classify new images accordingly. To improve transparency and trust, explainability methods such as Local Interpretable Model-Agnostic Explanations (LIME) and Shapley Additive Explanations (SHAP) are integrated to highlight the image regions that influence the model's predictions. These visual explanations help clinicians understand and verify the reasoning behind the results rather than relying solely on automated outputs. The system functions as a decision-support tool, assisting healthcare professionals in early screening and diagnosis, particularly in areas with limited access to specialists. By combining accurate prediction with clear interpretability, the system provides a reliable and practical approach for retinoblastoma detection.

1.6 Objectives

1. To design a deep learning–based system for early detection of retinoblastoma using retinal fundus images.
2. To design a classification model that categorizes fundus images into normal and retinoblastoma-affected classes with high accuracy.
3. To implement Explainable Artificial Intelligence (XAI) techniques such as LIME and SHAP to interpret and visualize the model’s predictions.
4. To develop a simple and user-friendly interface for uploading fundus images and displaying prediction results with visual explanations.

1.7 Methodology

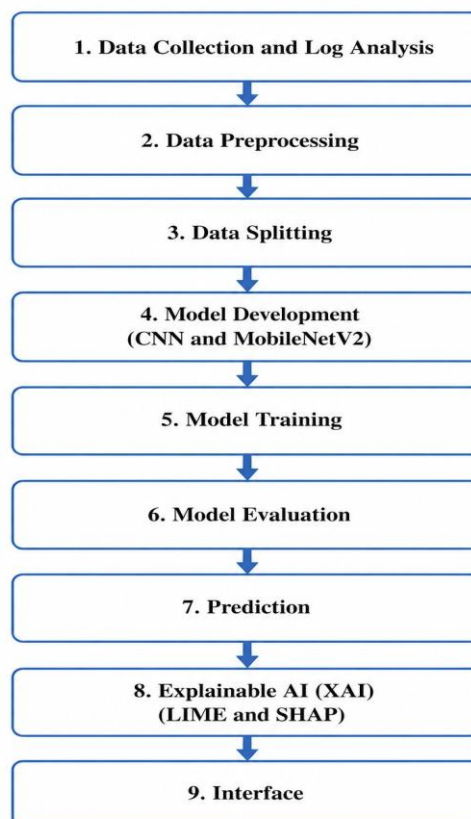


Figure 1.1 Methodology

Figure 1.1 shows the entire process of the Explainable Retinoblastoma Detection System for identifying retinoblastoma from retinal fundus images. The system uses deep learning and explainable AI to determine if the surveillance image is either normal or contains a tumor. The process consists of many different stages, beginning with data collection and closing with the final user interface that presents the predicted results and explanations.

The first stage of the process is Data Collection and Log Analysis, in which reliable data sources (medical databases and clinical records) are used to acquire retinal fundus images, which are both normal retinal images as well as images that have been affected by retinoblastoma tumors. The importance of accurately collecting data is essential since the quality and diversity of the dataset directly correlate to the model's performance. Additionally, conducting log analyses provides information about image distribution, class balance and overall quality of the data.

The second stage of the process is Data Preprocessing, in which the collected images are ready to be provided to the model for training and testing. In this stage, resizing images to a predetermined fixed resolution, normalization, removal of noise, enhancement of contrast and augmentation of the data via horizontal flipping and pixel adjustment will occur, resulting in enhanced quality of the image as well as increased effectiveness of the model's hierarchy of features. In the third step of machine learning, referred to as Data Splitting, the data set is divided into training, validation, and testing sets. The training set will be used to teach the model, the validation set will be used to monitor the model's performance while it is being trained, and the testing set will be used to measure the accuracy of the final model. The proper splitting of the data will allow for accurate and unbiased measurement of the performance of the final model.

In the fourth step of machine learning, referred to as Model Development, a Convolutional Neural Network (CNN) using the MobileNetV2 transfer learning approach is developed. The CNN allows for the extraction of features from the images (such as edges, shapes, and abnormal patterns), and the MobileNetV2 transfer learning allows for an efficient and effective method of predicting the model's output, based on an existing knowledge base of previously learned about images.

In the fifth step of machine learning, referred to as Model Training, the model learns the patterns from the training data set over a number of epochs. During this time, the system adjusts the weights of the CNN to minimize the loss associated with the models output so that the predictions are as accurate as possible. Training also allows the model to recognize the difference between a normal and an abnormal (tumor-affected) retinal image.

In the sixth step of machine learning, referred to as Model Evaluation, the trained model is being tested against a series of performance metrics including accuracy, loss, precision, recall, and confusion matrix. Performing this step will allow us to determine how well the model will perform on a new (real-world) prediction task. Step Seven is the Prediction step; Here at this stage the user will upload an image of the retina for analysis and prediction by the model. This stage allows the system to determine if the image is normal or abnormal and provides a confidence score in the prediction.

Step Eight is Explainable AI (XAI); At this stage the model uses LIME and SHAP to provide an explanation for its prediction. This is done by highlighting on the retinal image the specific areas that affected the model's decision to assist users and healthcare professionals in understanding how it arrived at its prediction.

Step Nine is the Interface; The web application has been developed using Flask, HTML, CSS and JavaScript. The web application provides a user-friendly interface in order to upload images, make predictions and view explanations clearly and as such is fully integrated into the overall workflow for the accurate detection, transparently providing information to end users, and filling a need for usability within the diagnosis of retinoblastoma.

1.8 Organization of the Project

Chapter 1 introduces retinoblastoma detection and provides some insight into the importance of early diagnosis, especially for children. It also discusses the challenges associated with diagnosing retinoblastoma in real life. The introductory chapter includes the problem statement, objectives of the project, and an overall overview of the proposed system. Chapter 2 will detail how retinal fundus images were obtained from participants and prepared for use in the model. It will also describe linear and non-linear preprocessing steps that were carried out prior to using the linear/non-linear data analysis models on these retinal fundus images. Chapter 3 will outline the dataset that was used in this study, and the critical role that accurately labelled images play in developing a valid retinoblastoma classification model. Furthermore, in order for the model to effectively classify/diagnose retinoblastoma from fundus images. The focus of Chapter 4 is the implementation stage of the project. It outlines how the model is trained and tested by utilizing the dataset. In addition, it discusses the evaluation process through the use of such metrics as accuracy, precision, recall and F1-score for assessing how well the model performed. In Chapter 5, we outline both the final results of implementing the system and how the system works. Within this chapter, we provide examples of how the model performs its predictions as well as how we generate explanation maps from our model predictions using XAI methods. Therefore, we also describe the process of developing a simple graphical user interface (GUI) in order to allow users to upload an image and subsequently view the model's prediction on that image along with the area(s) that were crucial in making the prediction. Lastly, the project concludes with a discussion regarding the overall performance of the system and recommendations for improving the system through higher accuracy and utilizing greater datasets to achieve better performance.

2. LITERATURE SURVEY

Researchers at Stanford University (Andre Esteva et al.) developed a convolutional neural network (CNN)-based deep learning algorithm for skin cancer detection that performs comparably to dermatologists on the skin cancer detection task. This finding underscores the ability of CNNs to autonomously learn complex visual characteristics from digital images and not require the manual extraction of such traits as previously required. The model trained on a very large dataset of labelled images yielded a very high accuracy and reliability of the CNN in skin cancer detection. Also, this work illustrates that CNNs have the potential to support the use of deep learning in detecting disease and increasing the availability of dermatology diagnostics. Nonetheless, large-scale annotated datasets and generalisation ability pose future challenges to the ability of the model to assist patients who were not included in the training dataset. [1]

A comprehensive outline was prepared by David H Abramson, and co-authors of the clinical aspects of retinoblastoma, including its clinical presentation, methods of diagnosis, and available treatment methods. Retinoblastoma is a rare but very aggressive form of cancer of the eye and the most common malignant tumor affecting children under the age of 10. Although retinoblastoma can usually be diagnosed with one or more methods, including a clinical exam and/or fundus photograph (or both), it is vital for physicians to know how the disease behaves and what their timelines should be when diagnosing and/or treating a child with retinoblastoma since most patients have to diagnose their own retinoblastoma before their physicians do. The authors do not use artificial intelligence in their research but may serve as rounded reports based on clinical characteristics of retinoblastoma, which will help to develop future artificial intelligence-enabled diagnostic tools. [2]

Dr. Eric J. Topol defines high-performance medicine as an approach to medical care that combines AI with clinical skill or expertise in order to increase the accuracy of diagnoses, relieve physician workloads, and enhance the overall efficiency of the health care system. While TB1.4.2.4 states that medical professionals will face challenges related to data ethics, privacy, and access to AI-enabled technologies, there are also significant regulatory difficulties associated with establishing rules and guidelines for the implementation of AI-based technologies into the practice of medicine [3].

The DenseNet Architecture presented by Gao Huang and colleagues combines all previous layers in a feedforward design so that they all have connections to one another. This design means that the features produced by a previous layer are reused at every other layer thus boosting the gradients through the network and Leading to a network that can be trained efficiently with a minimum number of parameters. Overall, this has produced large improvements in image classification tasks, particularly in medical image classification tasks. However, the dense connectivity of this architecture can lead to increased memory requirements which makes it difficult to deploy in resource limited settings [4].

Geert Litjens and others performed an extensive survey of current deep learning techniques for analysis of medical images; for all areas including the detection, segmentation and classification of disease. They showed how successful Convolutional Neural Networks (CNNs) are to extract meaningful features from medical images and improve the accuracy of diagnosis. Additionally, they describe numerous challenges associated with these techniques and AI (Artificial Intelligence) including limited annotated datasets available to train models, difficulties associated with interpreting AI results, and difficulties connecting the current state of AI functionally to existing clinical workflow. This work is an excellent and valuable source of information to understand the current state and future direction of the application of AI to medical imaging [5].

Kaiming He and his team have developed an enhancement of the original ResNet architecture by incorporating residual learning principles to overcome some of the challenges created by vanishing gradients in deep network architectures. The use of skip connections, or shortcuts, among layers supports training of much larger and more depth-oriented architectures than would otherwise be feasible. This enhancement leads to significant performance improvements for image classification tasks. The utility of this new architecture has made it a common choice amongst researchers performing analyses on medical images, since it has the capability to extract many complex patterns from the data; however, it also creates substantial additional computation requirements and longer required training time for larger models. [6]

The VGG network was developed by Andrew Zisserman and Karen Simonyan, featuring a straightforward and consistent design made up of deep layers and very small convolution filters. The VGG Network has been shown to behave quite well in image classification, and it has been utilized extensively as a reference point across a number of applications. However, the VGG Network's performance hence relies heavily on both processing power and memory, limiting potential applications in either real-time or low resource settings [7].

Marco Tulio Ribeiro et al. developed LIME, which is a methodology for interpreting machine learning model predictions. LIME works by creating locally approximating simpler, interpretable models to represent the predictions made by complex models. This provides a means of understanding how predictions are determined for complex models, which is particularly critical within the healthcare industry for establishing both trust and adoption of machine learning models. However, LIME may produce differing interpretations for different inputs, and the local nature of LIME does not always give a complete representation of the overall model [8].

Chexnet, a deep-learning algorithm developed by Pranav Rajpurkar and his team, is an example of how deep-learning models can be used to identify pneumonia using chest X-ray images with an accuracy comparable to radiologists. Chexnet also demonstrates the scalability of deep-learning models for other imaging applications in both medicine and elsewhere (e.g., ophthalmology). One limitation of Chexnet and other deep-learning models is that their success is very dependent on the quality and diversity of the training data that was used to build them [9].

Grad-Cam is a model introduced by Ramprasaath R. Selvaraju and other researchers that provides visualizations of which portions of an image contribute most significantly towards achieving a particular prediction from deep-learning models. Grad-Cam helps clinicians verify the output of AI-generated image analyses and increases their understanding of the algorithms used to generate these outputs by providing visual representations of how different parts of the input data were processed by the AI. Due to the complexity inherent in generating complex medical images, Grad-Cam may not always provide accurate representations of the input image [10].

Scott Lundberg and Su-In Lee developed SHAP, an integrated framework for interpreting predictions made by machine learning algorithms using Shapley values, which come from game theory. SHAP provides consistent and theoretically grounded types of explanations for predictions of machine learning models and therefore adds significant value in terms of improving the transparency of models. Nonetheless, SHAP can be computationally intensive, particularly for large and complicated models [11].

Varun Gulshan et al. designed and implemented a deep learning algorithm to detect diabetic retinopathy from images taken of the retina (fundus of the eye). This algorithm has high sensitivity and specificity, which makes it well-suited for use in population-based screening programs. Early detection of diabetic retinopathy is vital to avoid the loss of vision due to this disorder. Subsequent advances made by Daniel S. W. Ting et al. have demonstrated that this technique (i.e., detecting eye disease using deep learning algorithms and fundus photography) is also valid for detecting multiple eye diseases (coexisting diseases), verifying that AI has an important place in improving the practice of ophthalmology. However, all of these deep learning systems will encounter challenges in terms of the variability of datasets and generalizability across patient populations [12][15].

By introducing important principles of Deep Learning including; Neural Networks, Back Propagation and Representation Learning, Yann LeCun, Yoshua Bengio and Geoffrey Hinton helped to create the foundation for current advances in artificial intelligence and, in turn, have enabled the development of sophisticated AI models being applied in many different fields (including Medical Imaging). The ongoing development of AI continues to raise a number of significant research issues which include Data Dependencies, Model Interpretability and Ethical Considerations [13].

The Inception architecture was proposed by Christian Szegedy et al. and uses parallel Convolutional Filters of multiple sizes to create a more computationally efficient architecture. This technique creates an architecture capable of capturing multi-scale features and therefore improves the overall performance when performing an image classification task. However, the high level of complexity of the architecture can make implementation and optimization of the model cumbersome [14].

3. SYSTEM / MODULE DESIGN

The design of the System and the way the different components of the project work together are described in this chapter. Instead of working on the entire System at once, the System is divided into a number of Modules. Each Module has its own specific purpose and all the Modules come together to form the complete Process of detecting Retinoblastoma in the Retinal Image.

The design is kept simple to enable easy understanding and implementation by the end-user. The flow of information begins with receiving the input image and completes with displaying the output with explanation using the GUI.

3.1 Architecture of the System

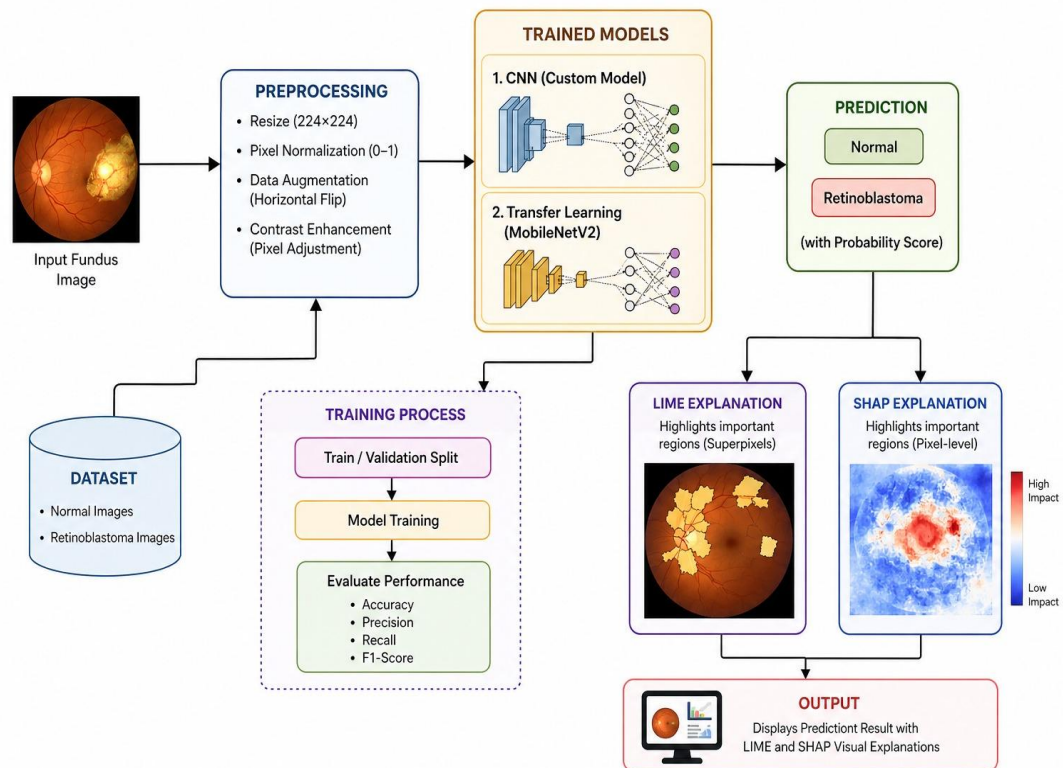


Figure 3.1 Architecture of the System

Figure 3.1 shows system architecture comprises a series of sequentially performed functions where each function has had its own specific purpose as well as producing outputs that subsequently become inputs into other functions. The initial point of entry into the system will be through allowing the user to upload retinal fundus images via a graphical user interface (GUI), thus making the system accessible and user-friendly. Once the user has uploaded the image, it will be sent to the pre-processing stage.

In the pre-processing stage of the process, the image will be resized to a constant resultant size to conform to the requirement necessary for model input; also, pixel values will be normalised, which would assist with appropriate model performance; and, where practical, simple techniques such as the removal of noise and improving image's quality may be implemented, hence improving clarity. The next stage involves using the previously trained deep learning model to process and analyse the uploaded image, with respect to identifying several key features (for instance, how well the model is able to identify whether abnormal patterns occur within one or more specific areas of the retina); the model will generate a classification based on analysis of each image passed as input into the system.

A pipeline-style series of defined stages defines the overall system architecture; each stage performs a distinct function and sends its output to the next stage. This pipeline-based construction allows for smooth data flow, providing modularity facilitating understanding of all data in use. The main purpose for developing this system was to identify abnormalities in retinal fundus images with an emphasis on using deep learning to identify retinoblastoma.

The first stage of the system consists of a Graphical User Interface (GUI), with users accessing through an easy to navigate and utilize interface; any user, even those that do not have an education in technology i.e., medical practitioner, will be able to access and use the interface without issue. Users will utilize the GUI to upload retinal fundus images in common formats (JPG/PNG); prior to processing the selected image will display within the GUI for the user to confirm it was uploaded correctly, helping to reduce the probability of a user-uploading an incorrectly respectively image that ultimately ends up being displayed as an abnormality. This particular stage additionally provides for enhanced accessibility to the user and creates a continuum of interaction between the GUI and the user.

The initial step is uploading an image which then is pre-processed to ensure the input data has good quality consistent with what is needed for reinforcement learning (AL). Pre-processing has two major functions of ensuring that the input is converted into a suitable size for the AI model being developed and ensuring pixel values within all images are scaled similarly; both of which are essential for efficient functioning of the AI model during prediction and increasing the consistency and effectiveness of the AI models across many different datasets. The pre-processing also reduces any potential noise or disturbances within each image by applying Gaussian blur and optionally enhancing certain features of the images such as the brightness and contrast so that the AI model can discern various features within the retina accurately. Finally, the result of the pre-processing step will be images that have been made clean, have been standardized in size, and are appropriate for performing an accurate analysis of retinas.

During training, the data augmentation process is applied to increase the number of variations and diversity of images that can be fed into an AI model, without collecting more data. The most commonly used data augmentation techniques include rotation, zooming in/out of the image; flipping the image horizontally; and making the image brighter and/or darker. The objective of this step is to provide the AI model with numerous examples of variations of a single image to improve AI model generalization, reduce overfitting due to an insufficient number of training samples available to create an accurate AI model, and assist the AI model in performing well with unseen images.

MobileNetV2 is used to extract features using the first few layers of the model, which will be used to identify features in the image(s) being evaluated (e.g., edges, texture, and abnormalities) so that the last few layers of the MobileNetV2 model can be modified and trained to classify whether the evaluated image(s) are either considered "normal" or "abnormal" (e.g., retinoblastoma). The features will pass through convolutional layers that will be used for feature extraction, pooling layers that will be used to reduce dimensionality, and fully connected layers that will make the classification. Once the features have passed through the convolutional, pooling, and fully connected layers, the final output of the MobileNetV2 model will be a probability score (0.0-1.0) that will indicate how likely an evaluated image is to have abnormal features.

Once processed by the model, each image is assigned a value in a continuous range from 0-1, representing its classification as either a normal or abnormal sample, with normal being closer to 1 and abnormal being closer to 0. The final classification of the test sample is based on this probability and is assigned according to a defined threshold value that creates clear and reliable decisions. Therefore, the output provided to the user as the final decision is a simplified version of the diagnostic result. To improve the reliability and interpretability of the system, Explainable AI (XAI) techniques have been integrated into the model. Examples of these XAI methods include LIME and SHAP, which help explain the model's reasoning by indicating which areas of an image contributed most significantly to the prediction; they also provide relative importance for each predictive feature. Consequently, these techniques help users to gain insight into how the model arrived at its decision; this understanding is especially valuable in the medical field, where users must trust the decisions made by the model.

Finally, the output will be displayed to the user through a graphical user interface (GUI). An example of information provided for a given image would be the predicted class (either normal or retinoblastoma) and a confidence score. In addition, if the explainable feature has been enabled, visual representation in the form of heatmaps will also be displayed. These outputs will provide the user with more information than just a prediction, while also providing an easier way for non-technical users to interpret them. The proposed system architecture is beneficial because it is easy to use, detects automatically, processes quickly, and is accurate. The use of transfer learning with the MobileNetV2 architecture will also make the system more efficient and minimize how much computing power it needs, which allows the system to operate on devices with very little available resources. The inclusion of explainable AI techniques within the system promotes trust and credibility in the results produced by the algorithms, and a modularly designed system can easily grow as additional functionality is developed for future use.

In the future, by adding real datasets from hospitals, this system can demonstrate increased accuracy and reliability. An increase in the size of the dataset used while also implementing more advanced deep learning models can result in even greater results from the algorithms. Lastly, including real-time detection capabilities, as well as deploying this system via mobile or cloud platforms, can maximize system accessibility.

This system architecture follows an organized and efficient flow from user input to ultimate prediction and explanation of results. Each stage of the system is necessary to ensure the output generated by the algorithms is reliable, accurate, and usable. This system utilizes the combined capabilities of transfer learning with MobileNetV2 and explainable AI to provide accurate predictions that are also transparent in their processes, making this system an achievable solution for use in today's healthcare systems.

3.2 Description of Modules

The Input Module serves as a means for importing a user's retinal fundus image via a graphical interface, or GUI. This is also where the Input Module serves as the primary interface of the entire system and it collects any retinal image that has been uploaded by a user and transmits the image to the preprocessing module before it is subsequently passed on to the next analysis step. It will check that the image uploaded has been uploaded correctly and is acceptable for processing in subsequent analysis steps.

The Pre-Processing Module acts as a converter and trainer for the retinal image in preparation for use in deep learning models. Pre-Processing activities help increase the quality and accuracy of the retinal images, so that deep learning models built may work optimally. Some common pre-processing activities performed are that the images have been resized to a standard size, pixel value normalization has occurred with all pixels set to be within the range of 0-255, and all images have been denoised or cleaned of any extraneous noise that may have impacted the pre-processed retinal fundus image. The end result of these activities is that the quality of the data that was inputted into the model has increased, which enhances the performance of the model.

Consisting of a trained deep learning model, such as a Convolutional Neural Network (CNN), the Model Module is the portion of the system that examines the pre-processed retinal image. Through the model developed on the data, the model will define characteristics of the pre-processed retinal image such as edges, texture, and other patterns found in the image. From the previously learned characteristics of the retinal images, models can then analyze whether the retinal fundus image appears to be free from disease.

This is an important module of the system. The Prediction Module is responsible for generating the final classification, through taking the output from the trained model and determining whether or not an image of the retina that is being assessed is normal or abnormal and the likelihood of being an indication of possible retinoblastoma. By utilizing all the predictions, this module produces the final prediction for a given clinical finding to assist both users and physicians with making better-informed decisions about patient management.

This module improves user trust in the system through increased transparency into how the model arrived at its predictions. The Explainability Module accomplishes this by showing users the important areas of the retinal image the model used to generate its prediction, using LIME and SHAP methodologies. By producing these explanation outputs, users will have a better understanding of why the model classified the image in a certain manner.

The GUI Module provides a simple and user-friendly means of accessing and using the system. This module enables the user to upload a retinal image, receive a prediction, and view the explanation outputs associated with a given prediction (i.e., LIME and SHAP). By providing both a user-friendly interface and a means of producing visualization outputs, both medical professionals and users with no background in technology are able to use the system easily and efficiently.

4. IMPLEMENTATION OF THE MODULES

4.1 Datasets used

For this project we used a dataset of retinal fundus images split into two classes Normal (healthy retina), Retinoblastoma (tumour affected retina). The dataset was divided into folders for both training and validation. A segment of the dataset was used to develop the model while the remainder was used for validation to examine performance. The dataset was pivotal in aiding the model to learn patterns related to the existence of a tumour.

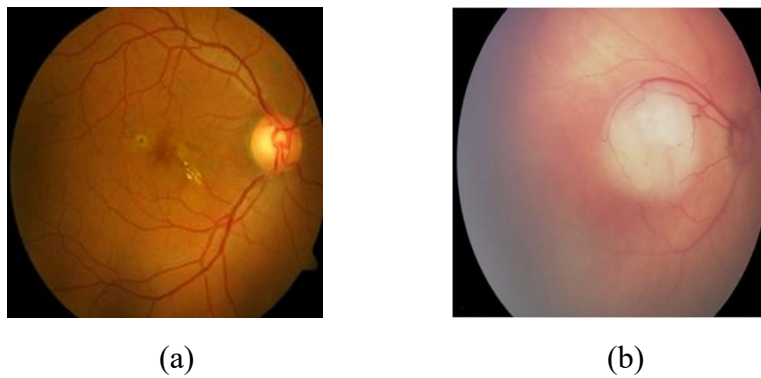


Figure 4.1 Fundus Images (a) Normal (b) Tumour

Figure 4.1 Fundus photographs illustrating the diagnosis of retinoblastoma. This is a normal retina (a) and one with a tumour (b), from the retina in the examples above. The retina in (a) is clearly a healthy retina and the retina in (b) has some prominent white reflections and other differences to the retinal structures seen in (a). These images show significant and easy to recognize differences and are useful to help the model to classify the two classifications as normal or abnormal accurately.

4.1.1 Data preparation

Pre-processing actions were conducted prior to training in order to maximize the ability of the model to perform and maintain consistency.

Image Resizing: Each image was re-sized to 224x224 pixels corresponding to the input size of the deep learning models.

Normalization: Pixel values were brought to a common numeric range of 0-1 in order to improve training stability.

Data Augmentation:

1. Horizontal Flips:

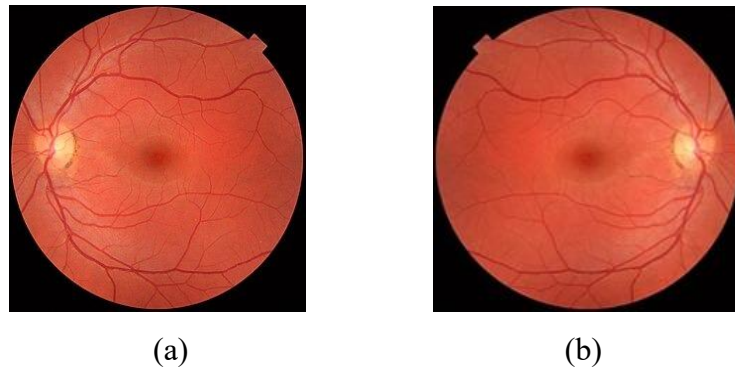


Figure 4.2 Images (a) Before Horizontal Flip (b) After Horizontal Flip

Figure 4.2 shows a horizontal flip transformation of a retinal photo. In (a) the original image is shown; in (b) the mirror image has been flipped horizontally on its vertical axis to change the left/right features of the image (left/right feature reversal). Flip transformation is commonly used for augmenting images with variations to improve the overall variety of the data set, while preserving the visual attributes of the original photographs.

2. Pixel Adjustments (Enhancing Contrast):

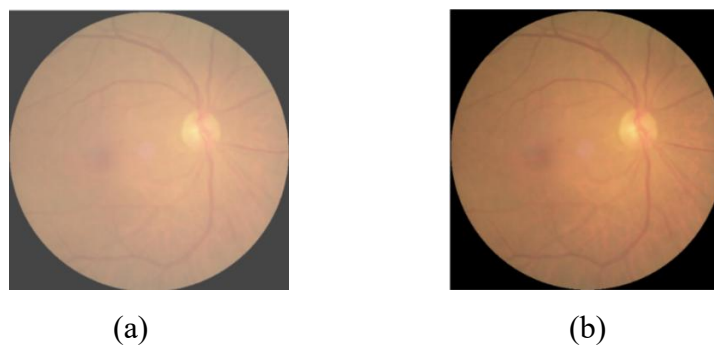


Figure 4.3 Image (a) Before Pixel Adjustment (b) After Pixel Adjustment

Figure 4.3 illustrates contrast in retinal images has been improved using pixel adjustment, as illustrated in the two images below. One image is an example of how limited contrast appeared to exist in the original image (a). The second image is the original image after pixel adjustment (b). It appears to have much better contrast and is easier to see the various structures in the retina.

3. **Dataset Structure:** Images were placed into directories (Normal / Tumour) to ensure proper training with data generators.

4.1.2 Training

The model was built using a binary classification based on the preprocessed images and then processed with Deep Learning techniques. This model used Image Classification methods and was created using the Red Ball algorithm and was optimized using the Adam optimizer with binary cross-entropy as a loss function. The model was trained over multiple epochs to help the model to learn. Class weights were employed to help to balance the disparity between the two classes. The model learned to find patterns of abnormality (i.e. areas that have irregularities in image), including abnormalities that appeared within the white spaces, irregularly shaped objects, Variations in image intensity levels.

4.2 Technologies used

Python is the programming language that has been used to build this application. TensorFlow/Keras, which is a deep learning application used to create and train deep learning ECM (Elastic Compute Model), is also part of developing and deploying the web-based application. The MobileNetV2 Convolutional Neural Network (CNN) allows you to classify images into categories (ex: dog vs cat) using image recognition techniques. OpenCV 4 is the open-source computer vision library that provides tools for uploading and pre-processing images (e.g., rescaling), and NumPy (short for Numerical Python) is an open-source programming language providing tools for manipulating and mathematical operations on image data. Matplotlib provides capabilities to generate and display graphs with model accuracy or results using the data output and classify the images. Flask is the web-backend framework used to provide the model into the web application. Lastly, HTML/CSS/JavaScript technologies have been used to create the front-end web application.

4.3 Results of CNN

The model utilized in this work was a Convolutional Neural Network (CNN) for the classification of retinal fundus images into normal and tumor categories. The CNN model achieved an accuracy of 69.97% during training and validation. The training accuracy improved gradually over the epochs, while the validation accuracy remained stable after minor fluctuations, indicating good learning capability of the model. The loss values decreased continuously throughout the training process, showing that the model was effectively optimizing and reducing prediction errors. The CNN model was able to extract important image features such as abnormal bright regions, irregular retinal patterns, and tumor-related structures from the fundus images. The preprocessing techniques and augmentation methods further improved the performance and helped reduce overfitting despite the limited dataset size. The model successfully identified most tumor images correctly and provided reliable classification results. Overall, the CNN-based approach demonstrated satisfactory performance for automated retinoblastoma detection and proved to be effective for medical image classification tasks.

4.4 Results of MobileNetV2

The model utilized in Algorithm-2 was MobileNetV2, a lightweight and efficient deep learning architecture suitable for medical image classification tasks. The MobileNetV2 model achieved an overall accuracy of 94.36% in classifying retinal fundus images into normal and tumour categories. During the training process, the training accuracy increased gradually across epochs, while the validation accuracy remained stable with minor fluctuations, indicating effective learning and reduced overfitting. The preprocessing techniques such as resizing, noise reduction, normalization, and data augmentation significantly improved the performance of the model. MobileNetV2 was able to extract important retinal features including abnormal bright regions, irregular retinal structures, and tumour-related patterns from the images. Compared to traditional CNN models, MobileNetV2 required fewer computational resources while still maintaining high classification accuracy. The model demonstrated reliable and consistent performance during testing and was effective in identifying retinoblastoma-affected images. Overall, the MobileNetV2 algorithm proved to be an efficient and accurate approach for automated retinoblastoma detection using retinal fundus images.

4.5 Comparisons of Algorithms

Table 4.1: Comparison of Model Performance Based on Accuracy

Model	Accuracy (%)	Performance Observation
CNN	69.97	Stable learning performance
MobileNetV2	94.36	Highest Performing

Table 4.1 compares the accuracy of different deep learning models used for retinoblastoma detection. MobileNetV2 achieved the highest accuracy of 94.36%, making it the best-performing model, while CNN showed the lowest accuracy.

5. RESULTS AND DISCUSSIONS

5.1 Results of each Module

The complete solution contains multiple modules and developed modules that output to lead into the following results:

Image Input module has successfully accepted retinal fundus images from the user via the provided interface and then utilised for subsequent processing modules.

Image Pre Processing Module each input image passed through the pre-processing image module underwent a series of processes including resizing the images to a single standard size of 224 x 224 pixels and pixel adjustment techniques to normalize and enhance overall image quality giving the images standardized size/quality thus allowing for better model performance.

Model Module the prepared training dataset was used to train two different models CNN and Transfer Learning model (v2). Both models were able to identify important feature structures from multiple retinal colour images and to distinguish between normal Vs retinoblastoma cases. The trained model was able to classify the images input into either category of Normal or Retinoblastoma.

The predictions were able to be generated in real-time. The LIME and SHAP techniques provided visual explanations indicating which areas in the retinal images influenced the model's decision. The visual explanation clearly indicated which areas of the retinal images were key factors influencing the model's decisions.

The graphical interface allowed users to upload images, should they wish to do so, and view the corresponding results and explanation maps, thus making the system user-friendly and encouraging user interaction.

5.2 Performance Measures

The performance of the system was evaluated using standard classification metrics for the reliable detection of Retinoblastoma from retinal images, including accuracy, precision, recall, and the F1 score. Accuracy describes the overall correctness of the model by calculating the number of correctly predicted images divided by the total number of images in the test set. Precision is a measure of the reliability of the model it is calculated as the ratio of images predicted as Retinoblastoma that were actually Retinoblastoma. Recall measures how well the model correctly identifies actual cases of Retinoblastoma. Recall is particularly important in medical diagnosis, where the failure to identify a case of disease can have a more serious impact than identifying a case of disease that does not exist. The F1 score is a valid measure of the overall performance of the model because it combines the precision and recall measures into a single performance metric.

The results indicate that this model has achieved high levels of accuracy along with acceptable levels of accuracy, precision, and recall, which suggests that it is both an accurate model and effective. The additional use of preprocessing techniques, in combination with the CNN model and the Transfer Learning V2 model for feature extraction and classification of retinal images, significantly enhances the overall performance of the system.

5.3 Comparative Analysis of Results

The findings from comparing the performance of the CNN model and Transfer Learning V2 for detecting retinoblastoma from retinal fundus images confirm the superior performance of the Transfer Learning V2 model due to its higher level of accuracy, precision, recall and F1 score in addition to the analysis of training and validation loss graphs. Specifically, the pre-trained weights used on the Transfer Learning V2 model were able to extract richer and deeper features from the images leading to improvements in accuracy, while the CNN model performed well but was limited by having to learn these features from scratch.

The loss graphs that were recorded for both models during training indicate that both models have been learning but that the Transfer Learning V2 model was much more stable and converged smoother than the CNN model. The accuracy graphs confirm that although both models produced high and consistent accuracy at the time of evaluation, the CNN model exhibited slight fluctuations in accuracy over the training process. Overall, the results of this study demonstrate that using a combination of pre-processing techniques in conjunction with the Transfer Learning V2 model significantly enhance the performance and reliability of the classification of retinoblastoma from retinal fundus images.

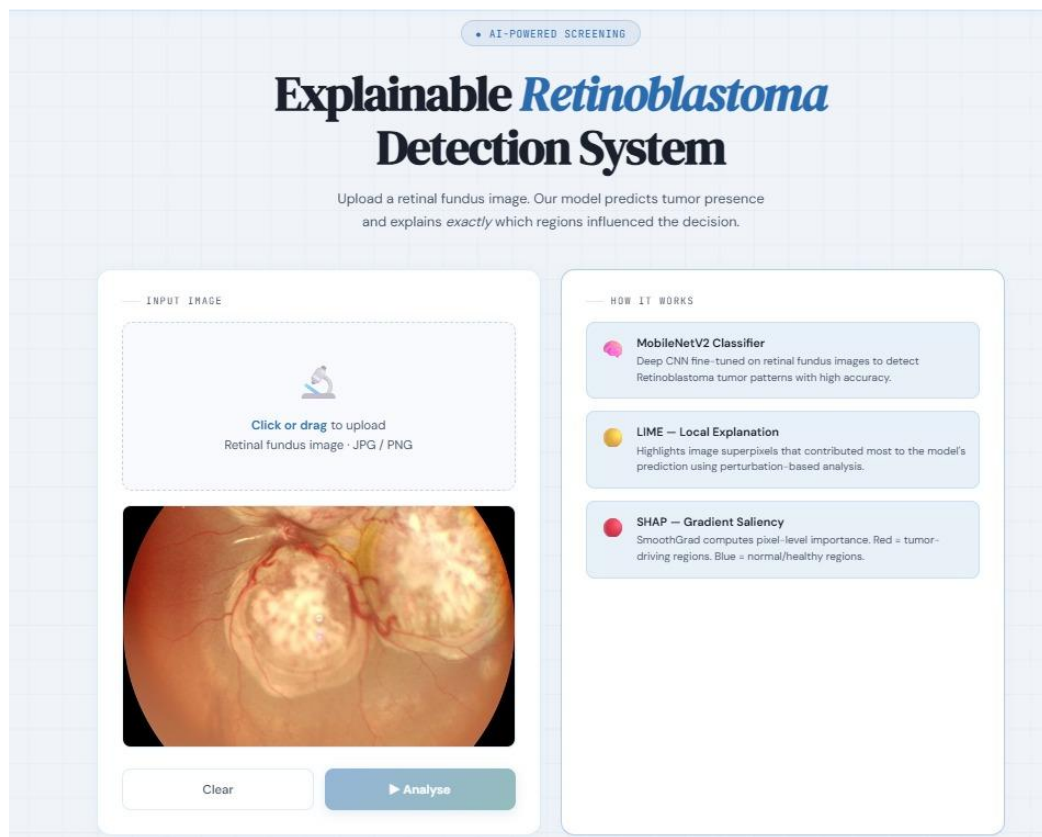


Figure 5.1 Interface

The Explainable Retinoblastoma Detection System is displayed in Figure 5.1. The interface allows the user to upload a retinal image for analysis against a trained model which will supply prediction results and explanatory information for each prediction. User will also be provided with information regarding the contribution of different regions in the image to the model's decision using techniques from LIME and SHAP to facilitate this process.

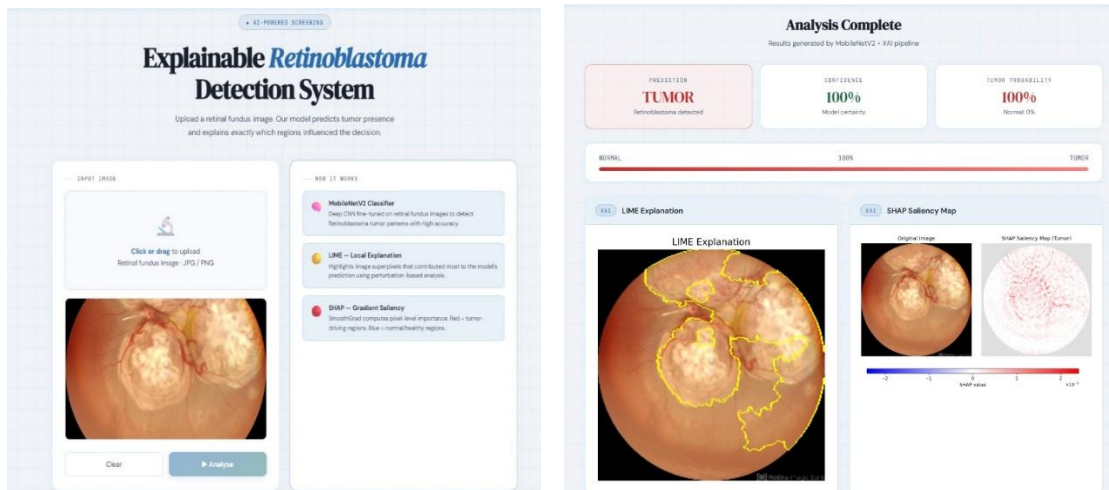


Figure 5.2 Tumour Detection

Figure 5.2 shows the results of the tumour detection analysis of the system. These include the identification of whether a tumour was detected in the retinal image uploaded and confidence scores associated with each of the predictions made by the model. The explanation methods (LIME and SHAP) also provide visual highlights to the parts of the image from which the predictions were made which assists users to better understand how the model predicted its predictions.

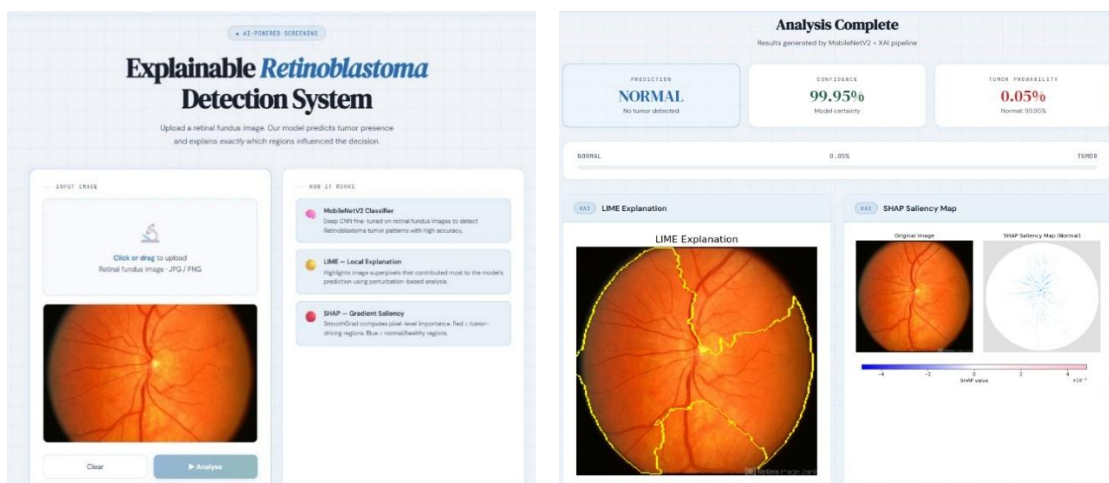


Figure 5.3 Normal Detection

Figure 5.3 shows the normal detection results from the system. The system can analyze and classify the retinal image as normal with a high confidence score indicating no tumour(s) is present. The areas of interest do not have abnormal patterns that can affect the detection of tumours, as shown by the LIME and SHAP visualizations.

6. CONCLUSIONS AND FUTURE ENHANCEMENTS

Conclusion

A successful AI-based system for detecting Retinoblastoma using retinal fundus images was developed in this project through the application of deep learning techniques. The project involved collecting and preprocessing retinal image datasets using methods such as resizing, normalization, noise reduction, contrast enhancement, and data augmentation to improve image quality and consistency for training. Deep learning models including CNN and MobileNetV2 were trained and evaluated, and the results showed that preprocessing and augmentation significantly improved model performance compared to using raw images alone. Among the implemented models, MobileNetV2 achieved the best performance due to its efficient feature extraction and classification capability. A user-friendly GUI application was also developed to allow users to upload retinal images and obtain real-time prediction results along with confidence scores. In addition, explainable AI techniques such as LIME and SHAP were incorporated to highlight the important regions of retinal images that influenced the model's prediction, thereby improving transparency and user trust in the system. Overall, the project demonstrated that deep learning combined with appropriate preprocessing and explainable AI techniques can effectively support automated Retinoblastoma detection from retinal fundus images

Future Enhancements

Although the present application has seen a lot of success, there still exists much more room for advancement. For increasing how well a model generalizes, need to make both larger datasets (more examples) and provide images that have greater diversity (differently). Both changes will help reduce how much a model overfits. The application must be capable of detecting other common forms of eye disease in addition to its current functionality. Such diseases include Diabetes Retinopathy, Glaucoma, and Age Related Macular Degeneration. To account for this enhancement, the existing binary class model must be developed into a multi-class model. The current model will benefit from being built upon a newer more advanced architecture (e.g., EfficientNet) in order to achieve greater model accuracy. Hyperparameter tuning must also be performed to reduce model training time without sacrificing accuracy even when reduced training times occur. The application must be deployed in the cloud using web service provisions to provide accessibility through use of a web address as opposed to requiring use of a local machine. There must be enhancements to the current deployment of the application that allow for scalability of the application so that many end-users can utilize it. A mobile app must be created so that end-users can utilize the new features of the app at their leisure.

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GLOSSARY

- **CNN:** A deep learning model used for image classification. It automatically detects important features such as edges, shapes, and patterns in images, making it highly effective for medical image analysis.
- **Retinoblastoma:** A type of cancer affecting the retina. It usually occurs in young children and can cause vision loss if not detected and treated at an early stage.
- **Fundus Image:** Image of the interior surface of the eye. It captures the retina, optic disc, macula, and blood vessels, helping doctors identify eye diseases.
- **Preprocessing:** Steps applied to prepare images before training. This includes resizing, normalization, noise removal, and enhancement to improve image quality and model performance.
- **Data Augmentation:** Technique to increase dataset size artificially. It involves rotating, flipping, zooming, and adjusting brightness of images to help the model generalize better.
- **Accuracy:** Percentage of correct predictions. It shows how well the model performs by comparing predicted results with actual labels.
- **Loss:** Measure of error during training. It indicates how far the model's predictions are from the actual values and helps in improving model learning.
- **LIME:** Technique to explain model predictions locally. It highlights specific parts of an image that influenced the model's decision for a single prediction.
- **SHAP:** Technique to show feature importance. It helps in understanding how much each feature contributes to the final prediction of the model.
- **Flask:** Framework used to build web applications. It is a lightweight Python framework that connects the trained model with the frontend for user interaction and prediction display.

APPENDIX

- **Convolutional Neural Network (CNN):** The Convolutional Neural Network is a deep learning method used for image classification and medical image analysis. It automatically identifies important features such as edges, textures, and abnormal regions in retinal fundus images, eliminating the need for manual feature extraction by specialists. Convolutional layers combined with pooling layers help the model learn complex image patterns, making CNN highly effective for retinoblastoma detection.
- **Local Interpretable Model-Agnostic Explanations (LIME):** LIME is an Explainable Artificial Intelligence (XAI) technique used to explain the predictions made by CNN models. It highlights the specific regions of the fundus image that most influenced the model's decision, helping doctors and users understand why an image was classified as normal or abnormal.
- **Shapley Additive Explanations (SHAP):** SHAP is another powerful XAI method used to improve model interpretability by assigning importance values to individual image features. It shows how much each feature contributes to the final prediction, increasing transparency, trust, and reliability in the model's output.
- **TensorFlow/Keras:** TensorFlow and Keras are deep learning frameworks used for building, training, and evaluating CNN models. They provide built-in functions for model optimization, image classification, and performance analysis, making the development process faster, easier, and more efficient.
- **OpenCV:** OpenCV is used for image preprocessing tasks such as resizing, normalization, noise reduction, and image enhancement. These preprocessing steps improve the quality of fundus images before they are given to the CNN model, which helps improve prediction accuracy.
- **Scikit-learn:** Scikit-learn is used for data splitting and performance evaluation. It helps calculate important performance metrics such as Accuracy, Precision, Recall, F1-Score, and Confusion Matrix, which are essential for measuring the effectiveness of the retinoblastoma detection system.